

1. (50 points) Calculate the Fourier transforms for the following signals:

$$x_1(t) = \delta(t-T)$$

$$\text{Fourier Transform: } X(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x_2(t) = \text{rect}\left(\frac{t}{\tau}\right)$$

$$x_3(t) = te^{-at}u(t)$$

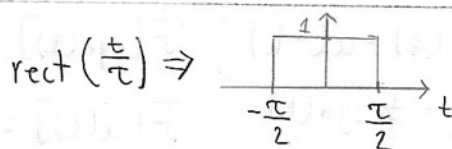
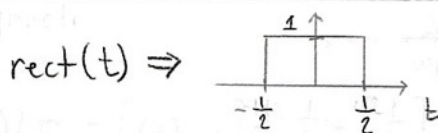
$$x_4(t) = \cos(\omega_0 t)u(t)$$

$$x_5(t) = e^{-at} \cos(\omega_0 t)u(t)$$

$$x_1(t) = \delta(t-T)$$

$$X_1(\omega) = \int_{-\infty}^{\infty} \delta(t-T) e^{-j\omega t} dt = e^{-j\omega T}, \text{ (use sifting property)}$$

$$x_2(t) = \text{rect}\left(\frac{t}{\tau}\right) \leftarrow \text{even symmetry}$$



$$X_2(\omega) = \int_{-\infty}^{\infty} \text{rect}\left(\frac{t}{\tau}\right) e^{-j\omega t} dt = \int_{-\tau/2}^{\tau/2} 1 \cdot e^{-j\omega t} dt = 2 \int_0^{\tau/2} \cos(\omega t) dt = 2 \left(\frac{\sin(\omega t)}{\omega} \right) \Big|_0^{\tau/2}$$

$$= 2 \left(\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\omega} - \frac{\sin(0)}{\omega} \right) = 2 \left(\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\omega} \right) = 2 \left(\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\omega} \cdot \frac{\tau/2}{\tau/2} \right) = \tau \left(\frac{\sin\left(\frac{\omega\tau}{2}\right)}{\frac{\omega\tau}{2}} \right)$$

$$= \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$X_2(\omega) = \tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$x_3(t) = te^{-at}u(t)$$

$$X_3(\omega) = \int_{-\infty}^{\infty} te^{-at}u(t) e^{-j\omega t} dt = \int_0^{\infty} te^{-at} e^{-j\omega t} dt = \int_0^{\infty} te^{-(a+j\omega)t} dt$$

$$u = t \quad dv = e^{-(a+j\omega)t} dt$$

$$du = dt \quad v = \frac{e^{-(a+j\omega)t}}{-(a+j\omega)}$$

$$= (t) \left(\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right) \Big|_0^{\infty} - \int_0^{\infty} \frac{e^{-(a+j\omega)t}}{-(a+j\omega)} dt = 0 - 0 - \left(\frac{e^{-(a+j\omega)t}}{(a+j\omega)^2} \right) \Big|_0^{\infty} = -\left(0 - \frac{1}{(a+j\omega)^2} \right)$$

$$= \frac{1}{(a+j\omega)^2}$$

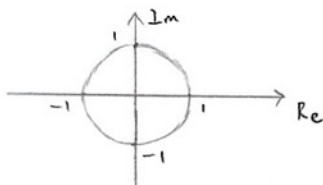
$$X_3 = \frac{1}{(a+j\omega)^2}$$

$$x_u(t) = \cos(\omega_0 t) u(t)$$

$$\begin{aligned} X_u(\omega) &= \int_{-\infty}^{\infty} \cos(\omega_0 t) u(t) e^{-j\omega t} dt = \int_0^{\infty} \cos(\omega_0 t) e^{-j\omega t} dt = \int_0^{\infty} \left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right) e^{-j\omega t} dt \\ &= \frac{1}{2} \int_0^{\infty} (e^{j(\omega_0 - \omega)t} + e^{-j(\omega_0 + \omega)t}) dt = \frac{1}{2} \left(\underbrace{\frac{e^{j(\omega_0 - \omega)t}}{\omega_0 - \omega} - \frac{e^{-j(\omega_0 + \omega)t}}{\omega_0 + \omega}}_{\text{cannot evaluate}} \right) \Big|_0^{\infty} \end{aligned}$$

$$e^{-j\omega t} = \cos(\omega t) - j\sin(\omega t) \Rightarrow$$

As t increases, $e^{-j\omega t}$ moves in a perfect circle.



Thus, $e^{-j\omega t}$ as $t \rightarrow \infty$ does not converge.

$$\text{sgn}(t) = u(t) - u(-t), \quad \mathcal{F}[\text{sgn}(t)] = \frac{2}{j\omega}$$

$$u(t) = \frac{1}{2} + \frac{1}{2} \text{sgn}(t), \quad \mathcal{F}[u(t)] = \mathcal{F}\left[\frac{1}{2}\right] + \frac{1}{2} \mathcal{F}[\text{sgn}(t)] = \pi \delta(\omega) + \frac{1}{j\omega}$$

$$x_u(t) = \cos(\omega_0 t) u(t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} u(t) = \frac{1}{2} (e^{j\omega_0 t} u(t) + e^{-j\omega_0 t} u(t))$$

Use frequency shift property, $\mathcal{F}[e^{j\omega_0 t} x(t)] = X(\omega - \omega_0)$

$$\begin{aligned} X_u(\omega) &= \mathcal{F}[x_u(t)] = \mathcal{F}\left[\frac{1}{2} (e^{j\omega_0 t} u(t) + e^{-j\omega_0 t} u(t))\right] = \frac{1}{2} \mathcal{F}[e^{j\omega_0 t} u(t) + e^{-j\omega_0 t} u(t)] \\ &= \frac{1}{2} \left(\pi \delta(\omega - \omega_0) + \frac{1}{j(\omega - \omega_0)} + \pi \delta(\omega + \omega_0) + \frac{1}{j(\omega + \omega_0)} \right) = \frac{\pi}{2} \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right) + \frac{1}{2j(\omega - \omega_0)} + \frac{1}{2j(\omega + \omega_0)} \\ &= \frac{\pi}{2} \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right) + \frac{2\omega}{2j(\omega^2 - \omega_0^2)} = \frac{\pi}{2} \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right) + \frac{\omega}{j(\omega^2 - \omega_0^2)} \\ &= \frac{\pi}{2} \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right) + \frac{\omega}{j(\omega^2 - \omega_0^2)} \cdot \frac{j}{j} = \frac{\pi}{2} \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right) + \frac{j\omega}{-1(\omega^2 - \omega_0^2)} \\ &= \frac{\pi}{2} \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right) + \frac{j\omega}{\omega_0^2 - \omega^2} \end{aligned}$$

$$X_u(\omega) = \frac{\pi}{2} \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right) + \frac{j\omega}{\omega_0^2 - \omega^2}$$

$$x_5(t) = e^{-at} \cos(\omega_0 t) u(t)$$

$$x_5(t) = e^{-at} \cos(\omega_0 t) u(t) = e^{-at} \left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right) u(t) = \frac{1}{2} (e^{-(a-j\omega_0)t} + e^{-(a+j\omega_0)t}) u(t)$$

$$X_5(\omega) = \mathcal{F}[x_5(t)] = \mathcal{F}\left[\frac{1}{2} (e^{-(a-j\omega_0)t} + e^{-(a+j\omega_0)t}) u(t)\right] = \frac{1}{2} \mathcal{F}[(e^{-(a-j\omega_0)t} + e^{-(a+j\omega_0)t}) u(t)]$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} (e^{-(a-j\omega_0)t} + e^{-(a+j\omega_0)t}) u(t) e^{-j\omega t} dt = \frac{1}{2} \int_0^{\infty} (e^{-(a-j(\omega_0-\omega))t} + e^{-(a+j(\omega_0+\omega))t}) dt$$

$$= \frac{1}{2} \left(\frac{e^{-(a-j(\omega_0-\omega))t}}{-(a-j(\omega_0-\omega))} + \frac{e^{-(a+j(\omega_0+\omega))t}}{-(a+j(\omega_0+\omega))} \right) \Big|_0^{\infty} = \frac{1}{2} \left[0 + 0 - \left(\frac{1}{-(a-j(\omega_0-\omega))} + \frac{1}{-(a+j(\omega_0+\omega))} \right) \right]$$

$$= \frac{1}{2} \left(\frac{1}{a-j(\omega_0-\omega)} + \frac{1}{a+j(\omega_0+\omega)} \right) = \frac{1}{2} \left(\frac{1}{a-j\omega_0+j\omega} + \frac{1}{a+j\omega_0+j\omega} \right)$$

$$= \frac{1}{2} \left(\frac{1}{(a+j\omega)-j\omega_0} + \frac{1}{(a+j\omega)+j\omega_0} \right) = \frac{1}{2} \left(\frac{(a+j\omega)+j\omega_0+(a+j\omega)-j\omega_0}{(a+j\omega)^2 + \omega_0^2} \right) = \frac{1}{2} \left(\frac{2(a+j\omega)}{(a+j\omega)^2 + \omega_0^2} \right)$$

$$= \frac{a+j\omega}{(a+j\omega)^2 + \omega_0^2}$$

$$X_5(\omega) = \frac{a+j\omega}{(a+j\omega)^2 + \omega_0^2}$$

2. (50 points) Calculate the inverse Fourier transforms for the following signals:

$$\hat{X}_1(\omega) = 1$$

$$\hat{X}_2(\omega) = \pi\delta(\omega) + \frac{1}{j\omega}$$

$$\hat{X}_3(\omega) = \frac{1}{a - j\omega}$$

$$\hat{X}_4(\omega) = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

$$\hat{X}_5(\omega) = \frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$$

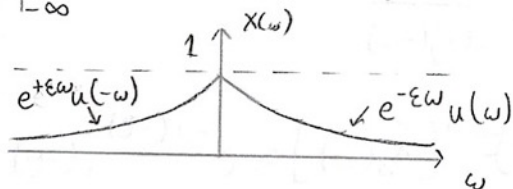
Inverse Fourier Transform

$$x(t) = \mathcal{F}^{-1}[\hat{F}(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{F}(\omega) e^{j\omega t} d\omega$$

$$\hat{X}_1(\omega) = 1$$

$$x_1(t) = \mathcal{F}^{-1}[\hat{X}_1(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} 1 \cdot e^{j\omega t} d\omega = \frac{1}{2\pi} \left(\frac{e^{j\omega t}}{j\omega} \right) \Big|_{-\infty}^{\infty} \leftarrow \text{does not converge}$$

$$\hat{X}_1(\omega) = 1 = \lim_{\epsilon \rightarrow 0} \chi(\omega), \quad \chi(\omega) = \begin{cases} e^{-\epsilon\omega}, & \omega \geq 0 \\ e^{+\epsilon\omega}, & \omega < 0 \end{cases}$$



$$x_1(t) = \lim_{\epsilon \rightarrow 0} \left[\frac{1}{2\pi} \int_{-\infty}^0 e^{\epsilon\omega} e^{j\omega t} d\omega + \frac{1}{2\pi} \int_0^{\infty} e^{-\epsilon\omega} e^{j\omega t} d\omega \right]$$

$$= \frac{1}{2\pi} \lim_{\epsilon \rightarrow 0} \left[\int_{-\infty}^0 e^{(\epsilon+jt)\omega} d\omega + \int_0^{\infty} e^{(jt-\epsilon)\omega} d\omega \right] = \frac{1}{2\pi} \lim_{\epsilon \rightarrow 0} \left[\frac{e^{(\epsilon+jt)\omega}}{\epsilon+jt} \Big|_{-\infty}^0 + \frac{e^{(jt-\epsilon)\omega}}{jt-\epsilon} \Big|_0^{\infty} \right]$$

$$= \frac{1}{2\pi} \lim_{\epsilon \rightarrow 0} \left[\frac{1}{\epsilon+jt} - 0 + 0 - \frac{1}{jt-\epsilon} \right] = \frac{1}{2\pi} \lim_{\epsilon \rightarrow 0} \left(\frac{1}{\epsilon+jt} + \frac{1}{\epsilon-jt} \right) = \frac{1}{2\pi} \lim_{\epsilon \rightarrow 0} \left(\frac{2\epsilon}{\epsilon^2 + t^2} \right)$$

$$= \frac{1}{\pi} \lim_{\epsilon \rightarrow 0} \left(\frac{\epsilon}{\epsilon^2 + t^2} \right) = \frac{1}{\pi} \cdot \pi \delta(t) = \delta(t)$$

$$x_1(t) = \delta(t)$$

$$\hat{X}_2(\omega) = \pi\delta(\omega) + \frac{1}{j\omega}$$

$$\mathcal{F}[1] = 2\pi\delta(\omega)$$

$$\mathcal{F}[\text{sgn}(t)] = \frac{2}{j\omega}$$

$$x_2(t) = \mathcal{F}^{-1}[\hat{X}_2(\omega)] = \mathcal{F}^{-1}\left[\pi\delta(\omega) + \frac{1}{j\omega}\right] = \mathcal{F}^{-1}[\pi\delta(\omega)] + \mathcal{F}^{-1}\left[\frac{1}{j\omega}\right]$$

$$= \frac{1}{2} \mathcal{F}^{-1}[2\pi\delta(\omega)] + \frac{1}{2} \mathcal{F}^{-1}\left[\frac{2}{j\omega}\right] = \frac{1}{2} (1) + \frac{1}{2} (\text{sgn}(t)) = \frac{1}{2} + \frac{1}{2} \text{sgn}(t)$$

$$= u(t)$$

$$x_2(t) = u(t)$$

$$\hat{X}_3(\omega) = \frac{1}{a-j\omega}$$

$$\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+j\omega}$$

Time scaling: $x(at) \leftrightarrow \frac{1}{|a|} \hat{x}\left(\frac{\omega}{a}\right)$

$$x(-t) \leftrightarrow \frac{1}{|-1|} \hat{x}\left(\frac{\omega}{-1}\right) = \hat{x}(-\omega)$$

$$x(-t) \leftrightarrow \hat{x}(-\omega)$$

$$\hat{X}_3(\omega) = \frac{1}{a+j(-\omega)}$$

$$x_3(t) = \mathcal{F}^{-1}[\hat{X}_3(\omega)] = e^{-a(-t)}u(-t) = e^{at}u(-t)$$

$$x_3(t) = e^{at}u(-t)$$

$$\hat{X}_4(\omega) = j\pi[\delta(\omega+\omega_0) - \delta(\omega-\omega_0)]$$

$$x_4(t) = \mathcal{F}^{-1}[\hat{X}_4(\omega)] = \mathcal{F}^{-1}[j\pi(\delta(\omega+\omega_0) - \delta(\omega-\omega_0))] = j\pi \mathcal{F}^{-1}[\delta(\omega+\omega_0) + \delta(\omega-\omega_0)]$$

$$= j\pi \cdot \frac{1}{2\pi} \int_{-\infty}^{\infty} (\delta(\omega+\omega_0) - \delta(\omega-\omega_0)) e^{j\omega t} d\omega = \frac{j}{2} \int_{-\infty}^{\infty} \delta(\omega+\omega_0) e^{j\omega t} d\omega - \frac{j}{2} \int_{-\infty}^{\infty} \delta(\omega-\omega_0) e^{j\omega t} d\omega$$

Use sifting property

$$= \frac{j}{2} (e^{j(-\omega_0)t}) - \frac{j}{2} (e^{j(\omega_0)t}) = \frac{j}{2} (e^{-j\omega_0 t} - e^{j\omega_0 t}) = \left(\frac{j}{2} \cdot \frac{j}{j}\right) (e^{-j\omega_0 t} - e^{j\omega_0 t})$$

$$= \frac{-1}{2j} (e^{-j\omega_0 t} - e^{j\omega_0 t}) = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} = \sin(\omega_0 t)$$

$$x_4(t) = \sin(\omega_0 t)$$

$$\hat{X}_5(\omega) = \frac{\omega_0}{(a+j\omega)^2 + \omega_0^2}$$

$$x_5(t) = \mathcal{F}^{-1}[X_5(\omega)] = \mathcal{F}^{-1}\left[\frac{\omega_0}{(a+j\omega)^2 + \omega_0^2}\right] = e^{-at} \sin(\omega_0 t) u(t)$$

$$x_5(t) = e^{-at} \sin(\omega_0 t) u(t)$$